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PURE MATH



The quantitative precision of algorithms makes them naturally suited to applications in mathematics. In this chapter, we explore algorithms that are useful in pure mathematics and look at how mathematical ideas can improve any of our algorithms. We'll start by discussing continued fractions, an austere topic that will take us to the dizzying heights of the infinite and give us the power to find order in chaos. We'll continue by discussing square roots, a more prosaic but arguably more useful topic. Finally, we'll discuss randomness, including the mathematics of randomness and some important algorithms that generate random numbers.